Documentation

Project Title: Navigation during Forestfires

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Start of the Project 10.03.2025

End of the Project: 23.03.2025

Date of Submission: 22.07.2025

Abstract:

The common wireless infrastructure requires cell towers which pose impractical or even hazardous in vast forest areas. This results in greatly reduced options when detecting forest fires in real-time and being able to efficiently navigate towards or away from them as firefighters or hikers respectively. Therefore, a device for navigation as the receiving counterpart to the Fire Detection Egg is proposed in this work.

Three keywords:

1. Navigation
2. Real-time fire detection
3. Digital prototyping

Resource Overview:

The current state of the project is a digital prototype. For this draft the following has been used:

- Visual Studio Code (as IDE)
- ChatGPT

In a physical prototype more components are necessary (see Document "Receiver Device")

1. Urgency / Context / Status Quo

How does ground navigation work in case of a fire? How can hikers be warned about fires if there is no cell service? How can firefighters navigate towards and away from fires?

Having reliable coverage of wireless communication access is difficult in a large forest. Not only is it not economically viable to build, maintain and service cell towers in almost-non-populated areas, but it could also lead to even more wildfires [1]. This makes access to real-time data about occurring wildfires in the area very difficult, if a person is already in the forest.

Monitoring active fires is also a challenge, because Satellites suffer from a trade-off between temporal and spatial resolution (either the Satellite can look at an area often or with precision) and are therefore relatively slow (about 2 days) [2].

There are efforts going on regarding Unmanned Aerial Systems (UAS) (e.g. drones), which have a significantly better resolution in both time and space, but are comparatively expensive, if you try to monitor big areas [2].

Another research point are Evacuation routes and AI-based modelling of wildfire-risks and several simulation approaches for active fire spread and risk assessment for different parts of the forest [3-5]. While they are valuable in prediction and intervention planning from a base with computational resources, it still does not offer hikers and firefighters on the ground the real-time information they might need.

For all these reasons, we identified the need for at least a supplementary option for real-time, inexpensive fire location tracking and navigation.

2. Topic / Approach / Solution

The goal of this idea is to design a device that is resource-friendly regarding electric components as well as battery usage and computing power. It is supposed to be robust against natural influences (i.e. fire, rain, extreme temperatures) which makes it durable.

The concept is a combination of a “Fire Detection Egg” and the corresponding “Navigation Receiver Device”.

In order to understand the design of the receiver (the protagonist of this work) let’s take a look at the sender:

As of now the sender is a multi-shelled ceramic egg. Ceramic is a natural and very robust material with a good temperature isolation that does not deteriorate with water and dirt. It is therefore perfectly suited for prolonged exposure outside where the Fire Detection Egg is placed and dormant.

Inside the outer shell are multiple Peltier-elements which convert a temperature difference into voltage. They act as the power supply in case of a wildfire, as there will be a temperature gradient due to the ceramic isolation between the (burning) outside world and the inside of the egg.

At the core is a Long Range (LoRa) transmitter transmitting the ID of the egg as a repeating message as soon as power is active. LoRa has the advantage of having a high range (~10km) and low energy consumption.

This is a great reduction in complicated electrical components as it does not require a charged power source – powered by the fire itself –
or sensors to detect the fire – only active in case of fire –
or even a GPS module – ID is sufficient.

Why is the ID sufficient? Because the Receiver Device does the work.

The Receiver Device is supposed to fill the gap of orientation and data access on foot for experts and laymen alike by functioning as a robust, handheld navigation device showing positions of fires nearby as indicated by the Eggs' signals. It is supposed to be affordable and intuitive to use.

This project focusses on creating a draft on how this device could look like and be interacted with.

The device aims to use rather cheap components that are easily available on the market such as simple GPS-modules or Magnetometers (digital compass). The desire is to have most of the data locally and only have wireless communication for the ID of the fire detection egg, which corresponds to a local database with GPS position entries.

It shows where you are in relation to the reported fires and overlays the contour lines of the area so you have an idea for efficient pathfinding.

3. Experiments / Steps of making process / Tools

The history of this project mainly comes down to finding a topic and finding the niche your strengths lay in.
The original idea was broadly "improving communication in case of fire" and started as a group project. We each had to figure out what made most sense for us in order to create something in two weeks.

So, what I decided on was to design something interactive as a digital prototype that would roughly outline the idea on how to interact with an analog counterpart so we could effectively communicate our idea for anyone who wants to continue this project.

There is always a huge barrier to starting any project if you think about everything that would have to work to move on to the next step. So instead, I used the assumption: "What if everything works? If all the technical components interact with each other the way we want them to, what would then happen with that data? How would the user interact with it?"

I chose Python and specifically Matplotlib to visualize a User Interface. It is not computationally efficient and in turn I also do not believe it is the best choice for further prototypes, but it is good for developing early drafts and visualizations.

I used ChatGPT for the rough skeleton of matplotlib interactive visualizations.
It did work for simple dragging movements for traversing the map, but I need to stress here, that the code snippets of most other specific functionality never worked as intended without modifications.
ChatGPT *did* make it faster to find the commands I could use to achieve a functionality. But without knowledge of the programming language – the ability to understand the code from

ChatGPT and identify the point I had to modify – it would not have worked with simple prompting.

I do not mean to discourage a motivated reader from realizing their projects with the help of AI. I believe it can be useful, especially when you have trouble starting from scratch, but as of now it does not replace knowledge of programming languages in general and needs to be used with caution and a high tolerance for frustration. (see the ChatGPT pictures)

I implemented different functionalities:

First, I needed the map based on the actual area. So, what is important about the area? I opted for contour lines of Corsica for now and downloaded them from https://www.opendem.info/download_contours.html. In future iterations it is desirable to also have rivers and pedestrian roads highlighted on the map similar to other Maps-Applications in their offline mode.

Second, I implemented conversions from the GPGGA format (the way GPS data is received) into longitude and latitude and computing a distance in kilometers as well as the angle relating to a compass direction in relation to my own location.
For the sake of having an early demo version, I wrote my own tiny database of ID and GPS-location pairs that could be randomly selected from in the main function.

Then, there was some tweaking in the representation of the arrows necessary. The arrow is red, if the fire is located in the depicted map part and it is yellow if it is outside of it.

Next to the interactive demo, I tried to think of all the technical parts necessary to enable the desired and useful functions for this kind of navigation device. When the time came to wrap the project up, I had a good grasp on my idea and could write it down for future iterations.

In conclusion, I attempted to sketch out the idea in text and computer visualization for future iterations. I got there through thinking through the needs of a potential user and some ChatGPT support.

4. Instruction to rebuild your final prototype

The visualization can be found in the "Code"-Folder which has a main function where you can select via "version" whether you want a specific sample of fire locations or random fire locations from a small sample data bank.

As part of my project is the description of the device itself, it is also added to this project folder (In English and in German).

5. Reflection

I think the navigation device could fill a gap where we are trying to use more high tech, expensive gear to map wildfires with UAS and have simulation tools for a base that plans the deployment of firefighters, but there is no device to take into the field to tell you where the fire is, especially if the situation suddenly changes and you do not have the financial capacity for several drones mapping your forests.
I tried to focus on low-cost and comparatively low-tech to form a first idea on what a device that fills this gap could look like.

I would have liked to include more functions and interface them with real sensors. I achieved a solid draft for the device specifying which sensors would be useful in the scenario and even if the code is largely redone in a different programming language I worked with the principle of modularity in mind with interfaces prepared for real data after it has been processed.

With more time I believe many more iterations of this project can be achieved.
The detailed description is (hopefully) now standalone enough that someone could understand the idea and requirements and do the next iteration.

Regarding technical skills, I took from this project how AI can help me in the early stages of a project. On a conceptual level, it was also a good atmosphere to let members of a group project figure out what they are most interested in and naturally find their role. Furthermore, because the project topic was not given, it was an interesting to go through the process of how to get from the emotion of “We have to do something about this” to a graspable idea.

References

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